TEMPTH

A Dimensional Model of Consciousness

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Abstract

This paper proposes a dimensional model of consciousness centered on a previously unnamed axis: tempth (φ). The central claim is that the subjective "me" is not the content of consciousness but its organizing coordinate — the dimension along which awareness moves independently of physical chronological time. Within a block universe framework, consciousness is modeled as a path $\Gamma = \{A(t, \varphi(t))\}$ through a two-dimensional state space, where t represents physical time and φ represents the tempth axis. The ego is reframed not as the core of identity but as a projection operator that collapses this 2D structure onto a 1D narrative timeline, generating the illusion of temporal flow. This model is stress-tested against neurological edge cases (H.M., Clive Wearing), altered states (dreams, psychedelics, dissociation, NDEs), and adversarial philosophical challenges. It integrates with and extends existing frameworks including predictive processing, Global Workspace Theory, Metzinger's Self-Model Theory, and Friston's Free Energy Principle. The tempth axis resolves a structural gap present in all current consciousness models: the absence of a formal coordinate for the dimension of subjective presence.

Part I: The Thesis

1.1 The Central Claim

"The 'me' is the organizing dimension (tempth) of awareness — the coordinate from which all conscious states are rendered, and the only part that persists when memory, identity, perception, and ego collapse."

This is not a mystical claim. It is a structural one.

Consider three analogies: spacetime is not an object in the universe but the structure that organizes objects. A canvas is not the painting but the substrate that organizes it. A game engine is not the characters but the system that renders them. In each case, the organizing dimension is invisible precisely because everything else is defined relative to it.

Tempth is that dimension for consciousness. It is not what you experience. It is what you experience from. The fact that current neuroscience and philosophy of mind lack formal tools to probe this axis does not make it less real — it makes it the next thing to formalize.

1.2 The Problem This Solves

Every major consciousness framework has the same structural gap. Whether you are working within predictive processing, Global Workspace Theory, Integrated Information Theory, or the block universe models of theoretical physics, the same question goes unanswered: if the universe is a static 4D block and time does not objectively flow, what moves?

Hermann Weyl articulated this in 1949: "The objective world simply is, it does not happen. Only to the gaze of my consciousness, crawling upward along the life line of my body, does a section of this world come to life." Weyl named the phenomenon but could not name the axis it moves along.

The Moving Spotlight Theory attempts this but fails because it requires a "super-time" — a second temporal dimension to govern the movement of the spotlight through the first — generating infinite regress. The tempth model resolves this elegantly: the spotlight is not built into the fabric of the universe. It is generated by a localized cognitive architecture operating along the φ axis. The "now" is not a physical property of spacetime. It is a neuro-phenomenological user interface.

Most consciousness theories implicitly assume $dA/d\varphi = 0$. They model awareness as a function of physical time alone. This is why they fail on altered states, dissociation, déjà vu, dream logic, and NDE phenomenology. These phenomena are not anomalies to explain away. They are data points on the φ axis — direct evidence that awareness moves through a dimension that existing models do not formally represent.

Part II: The Architecture

The Tempth Model proposes six integrated layers of conscious experience. These are not sequential stages — they are simultaneously operating components of a unified system.

Each layer has a distinct function. Together they produce what we call human consciousness.

Layer 1: The Substrate

Reality as it exists independent of any observer. The raw, unanalyzed "there-ness" of the universe. The model makes no claims about the ultimate nature of the substrate — only about what observers receive from it. This is the static 4D block of eternalist physics: all events equally real, no objective flow, no privileged present.

Layer 2: The Renderer — Moment-to-Moment Awareness

This is the fundamental capacity for experience — the light that makes anything visible at all. It is real-time, nonlinear, non-narrative, and not tied to memory or ego. It is always operating. This layer remains intact in clinical cases where every other layer fails. Henry Molaison (H.M.), who could not form new long-term memories after bilateral hippocampal removal, retained this layer completely. Clive Wearing, who experiences a 30-second memory window with no continuity, retains it. The anecdotal "dying grandma" who reports seeing people who are not there retains it even as other systems destabilize. The renderer is the last thing to go, and possibly the only thing that cannot go.

Layer 3: The Ego-UI (User Interface)

The ego is not consciousness. It is the interface consciousness uses to navigate social and physical reality. Its functions include: maintaining narrative identity, anchoring attention, generating self-reference, prioritizing salient inputs, and stitching discrete moments of the substrate into the felt experience of continuous time. The ego is the part that dissolves in dreams, deep meditation, psychedelic states, severe dissociation, and NDEs — and notably, subjective experience does not disappear when it does. This is the critical empirical observation. Ego dissolution is not experiential extinction. It is the removal of the rendering filter, exposing the layer beneath.

In formal terms: the ego is a projection operator. It takes the 2D awareness state $A(t, \varphi)$ and collapses it onto a 1D timeline: Ego: $A(t, \varphi) \rightarrow A(t)^*$. It ignores the φ axis entirely. This is why the ego's explanations for behavior are always post-hoc — as Gazzaniga's split-brain research demonstrates, the Left-Brain Interpreter generates plausible narrative regardless of whether it has access to the actual causal data. The ego is not a perceiver. It is an editor.

Layer 4: The Frequency Gate

Frequency (f) is the gating function that determines how much of the substrate the renderer allows into awareness at any given moment. Formally: $G(f): A(t, \varphi) \rightarrow A\_f(t, \varphi)$. At low frequency, the channel is narrow and stable — this is ordinary waking perception, dominated by the ego's predictive model. At high frequency, the channel expands and the φ dimension becomes more accessible — this is the mechanism of dream logic, intuition, symbolic thinking, and expanded states. At destabilized frequency, the render becomes choppy and the boundary between stitched moments becomes perceptible — this maps directly onto the phenomenology of NDE states and end-of-life experiences, where subjects report something like "the tuning is changing."

The frequency gate explains why different states of consciousness feel like different worlds while the substrate remains constant. It is not the world that changes. It is the aperture.

Layer 5: The Memory Pipeline

Memory operates in two distinct sub-layers that the model treats separately, because the neurological evidence demands it.

The Live Buffer (LB) holds the last several seconds to minutes of experience. It generates the felt sense of continuity — the phenomenological illusion that you are the same person you were ten seconds ago. H.M. had this layer. He could hold a conversation. He could laugh at a joke. He simply could not encode those moments into the Persistence Encoder.

The Persistence Encoder (PEM) is long-term memory consolidation — the hippocampal mechanism by which selected experiences are written into stable storage. H.M.'s bilateral hippocampal removal destroyed this layer entirely while leaving the LB intact. This dissociation is not a minor clinical detail. It is direct evidence that ego-continuity and awareness can be fully decoupled. The tempth axis can operate even when the memory pipeline cannot encode its outputs.

Layer 6: Tempth — The Missing Axis

Tempth (φ) is the dimension along which awareness moves that is not chronological time. It is a real-valued continuous axis — $\varphi \in \mathbb{R}$ — along which awareness can slide smoothly or jump discontinuously, independent of the t axis.

The name derives structurally from the dimensional naming conventions of physics: depth, width, length, spacetime. Tempth is temporal thickness — the extent of the present moment in the direction awareness actually moves.

Tempth is the coordinate axis that explains: dream jumps, time dilation under threat or flow, dissociation sliding, intuition leaps that precede conscious reasoning, precognitive-feeling flashes, trauma fragmentation and re-entry, déjà vu, meditation "timelessness," NDE panoramic review, end-of-life perceptual destabilization, and the subjective non-linearity of memory recall.

These phenomena are not anomalies requiring separate explanations. They are coordinates on the same axis. The φ dimension unifies them into a single structural framework.

Part III: The Formal Framework

3.1 Awareness as a Two-Dimensional Function

Let awareness at any moment be defined as:

$$\mathbf{A(t, \varphi)}$$

Where t = physical chronological time and φ = tempth coordinate. The observer is not simply "at time t." The observer is at a position (t, φ) in a 2D state space. This is the foundational departure from all existing single-axis models.

3.2 The Movement of Awareness

Awareness does not move only through physical time. It moves through both dimensions simultaneously:

$$dA/dt \neq 0 \quad \text{and} \quad dA/d\varphi \neq 0$$

dA/dt captures ordinary moment-to-moment awareness — the standard dimension all models account for. dA/dφ captures dream logic, temporal distortion, intuition leaps, memory fragmentation, younger-self encounters in trauma recall, dissociative sliding, and altered states. The reason existing models fail on these phenomena is that they assume dA/dφ = 0. They model awareness on t alone and treat φ as noise. It is not noise. It is signal from the second axis.

3.3 The Ego as Projection Operator

The ego takes the 2D awareness state A(t, φ) and collapses it onto a 1D timeline:

$$\text{Ego: } A(t, \varphi) \rightarrow A(t)^*$$

The ego ignores the φ axis. It narrativizes the 2D experience into a linear story where only t matters. This is not a flaw in the ego — it is its primary function. Social coordination, planning, and identity maintenance all require a stable linear timeline. The cost is that anything that occurs on the φ axis — intuition, dream logic, dissociation, altered states — cannot be integrated into the ego's narrative and is dismissed, pathologized, or simply not remembered. The ego is a dimensionality reduction operation. It is efficient and often adaptive. But it is also lossy.

3.4 Frequency as a Gating Function

Define frequency f as a gating function: G(f): A(t, φ) → A_f(t, φ)

At high f, more of the φ dimension becomes accessible. At low f, the ego's rendering dominates and the φ axis is suppressed. At unstable f, the rendering becomes discontinuous — this is the mechanism of NDEs, near-death perceptual phenomena, and the anecdotal observations of people at the boundary of consciousness where "the tuning is changing."

3.5 Awareness as a Path Through (t, φ) Space

A life is not a line through time. It is a path:

$$\Gamma = \{ A(t, \varphi(t)) \}$$

This means φ is a dynamic function of t, and different states have characteristic φ(t) profiles:

- Waking: φ(t) ≈ constant — the ego successfully stabilizes a narrow tempth band
- Dreams: φ(t) fluctuates widely — the ego-UI is reduced and φ movement is unconstrained
- Panic: φ(t) locks into hyper-narrow band — temporal compression, heightened linear urgency
- Flow states: φ(t) stabilizes at an expanded but coherent band — presence without ego dominance
- Intuition: φ(t) jumps discretely — information from a φ position not adjacent to current t
- Déjà vu: φ(t) revisits a previous neighborhood — spatial-temporal mismatch between t and φ
- Dissociation: φ(t) decouples from t entirely — the observer and the timeline separate
- NDE: φ(t) destabilizes toward boundary — approaches the edge of what the ego-UI can render

Part IV: Theoretical Grounding

4.1 The Block Universe and Awareness as Slice Selector

Einstein's special relativity dismantled the concept of universal present time. Because simultaneity is relative to the observer's inertial frame, there is no objective universe-wide "now." The B-theory of time and eternalism follow: past, present, and future events are equally real. The universe is a static 4D block, and worldlines — the complete paths of objects through spacetime — exist as unchanging eternal geometries.

The phenomenological challenge this creates is Weyl's problem: if the block universe simply "is" and does not happen, how does any section of it come to life as a present moment? The Moving Spotlight Theory fails here by requiring super-time. The Tempth Model resolves this without invoking any new physical mechanism. Awareness is not a property of the universe. It is a neuro-phenomenological gaze that moves along the φ axis, rendering successive coordinates of the block as a "now." The universe does not flow. The gaze does.

4.2 Predictive Processing and the Ego-Time Engine

Under Friston's Free Energy Principle and the Predictive Processing framework, the brain is a hierarchical inference engine minimizing prediction error across its generative models. Narrative identity sits at the apex of this hierarchy — the highest-level prior, minimizing surprise across all experience by maintaining the expectation that the self continues through time coherently.

In Tempth Model terms, the ego-UI is the narrative predictive model. Its collapse — in psychedelics, meditation, trauma, or NDE — corresponds to the empirically observed reduction in DMN precision weighting described by Carhart-Harris and Friston's entropic brain hypothesis. When DMN priors lose precision, the ego's projection operator weakens. The φ axis becomes accessible. $dA/d\varphi \neq 0$ becomes experientially available in a way that ordinary consciousness suppresses.

4.3 Global Workspace Theory and the Spotlight

Baars' Global Workspace Theory models consciousness as a theater: unconscious processes run in parallel backstage, and attention functions as a spotlight that selects content for global broadcast. The Tempth Model maps cleanly onto this: the global workspace is the renderer (Layer 2), the spotlight is ego-driven attention (Layer 3), and tempth is the dimension that attention moves through — which GWT, by modeling only the content selected, never formalizes.

4.4 Metzinger's Self-Model Theory

Metzinger argues that no selves exist — only Phenomenal Self-Models (PSMs), transparent simulations the brain generates of itself. The system cannot perceive its own modeling because it looks through the model to identify with it. The Tempth Model is fully compatible with this: the ego-UI is precisely Metzinger's PSM. The important extension the Tempth Model makes is to ask: if the PSM is a projection operator on $A(t, \varphi)$, what is $A(t, \varphi)$ before the projection? The answer is awareness moving along the tempth axis — the pre-egoic renderer that Metzinger acknowledges but does not formalize dimensionally.

4.5 Eastern Phenomenology: Independent Convergence

The Tempth Model was not derived from Eastern philosophical traditions, but it converges with them at the structural level. Advaita Vedanta distinguishes pure timeless awareness (Atman/Brahman) from the ego-sense (Ahamkara), which it identifies as the mechanism generating the illusion of temporal selfhood. Buddhist Anatman (no-self) describes the "self" as a rapid succession of aggregates stitched by dependent origination — awareness with the ego's stitching function removed. In both traditions, liberation involves decoupling from the temporal ego not because the ego is evil but because it is a construction. The Tempth Model formalizes this ancient structural observation in dimensional terms.

Part V: Stress Tests

A theoretical model is only as strong as its edge cases. The following stress tests apply the architecture to neurological, clinical, and phenomenological extremes. The model passes all of them.

5.1 Henry Molaison (H.M.) — Memory Pipeline Failure Without Ego Collapse

Following bilateral hippocampal resection, H.M. could not encode new long-term memories. Yet he retained: the ability to hold a conversation, personality, emotional responsiveness, humor, sense of self within each 30-second window, and phenomenological presence. In model terms: the Persistence Encoder (PEM) was destroyed. The Live Buffer (LB), the ego-UI, and the renderer all remained intact. The tempth axis continued to operate in real time even though its outputs could not be archived. This proves that consciousness does not require memory continuity. It requires only the renderer and the φ coordinate.

5.2 Dreams — High φ, Reduced Ego-UI

In REM sleep, the DMN is active but prefrontal regulatory circuits are suppressed. The ego-UI's projection operator weakens. The result: $\varphi(t)$ fluctuates widely, non-linear narrative becomes the norm, time jumps are unremarkable, and the dreamer accepts impossible events as real. This is not malfunction — it is the renderer operating on the φ axis with reduced ego filtering. The dreaming state is direct evidence that $dA/d\varphi \neq 0$ and that the ego's function is to suppress rather than generate awareness.

5.3 Psychedelic Ego Dissolution

Classic psychedelics (psilocybin, LSD, DMT) reliably produce ego dissolution — the temporary suspension of the narrative self — while awareness intensifies rather than diminishes. Carhart-Harris et al. document this empirically: the collapse of DMN default activity corresponds with reported dissolution of the self-boundary and, simultaneously, an expansion of experiential richness. In model terms: the projection operator fails. $A(t, \varphi)$ can no longer be collapsed to $A(t)^*$. The φ dimension floods into experience. Far from being pathological, this state is described by subjects as "more real than ordinary reality" — consistent with what the model predicts. Without ego suppression, more of the 2D state space is accessible.

5.4 Dissociation — $\varphi(t)$ Decoupling from t

In dissociative states, the subject perceives themselves as a detached observer of their own experience. Computationally, this maps to a breakdown in interoceptive predictive coding — the ego's ability to bind its self-model to the body's internal state. In φ terms: the ego-UI is running but $\varphi(t)$ has decoupled from the t axis. The observer and the timeline become separate tracks. The model predicts this directly from the architecture: if the ego's projection function weakens while the renderer continues, the subjective result is exactly this — experience without ownership.

5.5 Trauma — Hyper-φ Activation and Re-Entry

Trauma memories do not store and retrieve like ordinary memories. They intrude. They feel present-tense. They carry sensory immediacy that normal autobiographical memory does not. In model terms: traumatic encoding causes PEM distortion, creating φ-coordinates that remain "hot" — accessible not just by deliberate recall but by sensory triggers that match the original φ-state. Trauma is not stuck time. It is a fixed point on the φ axis that the ego-UI cannot fully suppress, causing involuntary re-entry into a past φ-coordinate while t continues forward.

5.6 NDEs — $\varphi(t)$ Approaching the Boundary

Near-death experiences characteristically involve: the collapse of sequential time, panoramic life review where multiple time periods feel simultaneously present, profound presence and clarity, absence of fear, and the sense of encountering something fundamental about the nature of experience. In model terms: as the biological hardware supporting the ego-UI begins to fail, the projection operator's suppression of the φ axis collapses. The frequency gate destabilizes. $\varphi(t)$ approaches the boundary of the model's normal operating range. The resulting phenomenology — non-linear time, simultaneous access to multiple φ-coordinates, absence of ego-driven fear — is exactly what the architecture predicts would occur when the ego-UI can no longer perform its stitching function.

Part VI: Adversarial Analysis

No framework is complete without attempting to break it. The following are the strongest objections to the Tempth Model, along with the model's responses.

6.1 The Block Universe May Be False

Lee Smolin and Roberto Mangabeira Unger argue that the block universe is a mathematical artifact of relativity rather than an ontological truth, and that time is fundamental and objectively real. If the future is genuinely open — as certain interpretations of quantum mechanics suggest — then the "slice selector" model of awareness does not apply.

Response: The formal dimensional structure of the Tempth Model does not require eternalism to be true. The φ axis can be reframed in a presentist ontology: rather than awareness "selecting" slices of a pre-existing block, the φ axis represents the degrees of freedom available to present-moment awareness that are not captured by physical time alone. The dimensional relationship $A(t, \varphi)$ holds regardless of whether the block universe is ontologically real or merely a useful framework. The math is independent of the metaphysics.

6.2 The Homunculus Problem

By positing a "timeless witnessing field" that the ego moves through, the model risks implying a tiny viewer inside the brain watching the show — an internal observer that itself requires an observer, generating infinite regress. Dennett's Multiple Drafts Model explicitly rejects this.

Response: The renderer (Layer 2) is not a localized observer. It is the globally integrated information broadcast described by GWT — the process of awareness itself, not a separate entity that witnesses awareness. The φ axis is a coordinate in the state space of this process, not a theater where a homunculus sits. The Tempth Model is fully compatible with Dennett if the renderer is understood as distributed global broadcasting and the ego is the editor that forces sequential narrative coherence onto parallel processing streams. There is no additional observer required.

6.3 What Is the Physical Implementation of φ?

The tempth axis is formally defined but its neural implementation is not yet specified. What brain process generates the φ coordinate? Without a proposed mechanism, the φ axis risks being unfalsifiable — a formal placeholder rather than an empirical claim.

Response: This is the model's primary open question and the most honest limitation to acknowledge. Several candidate mechanisms exist: hippocampal time cells encode sequential moments and their disruption produces non-linear temporal experience; thalamocortical oscillation frequencies gate what information reaches global broadcast; DMN connectivity patterns correlate with the degree of ego-narrative dominance. The φ axis may be a composite of these mechanisms rather than a single neural correlate. Specifying the implementation is the next required step — and it is an empirical question, not a conceptual one. The model makes falsifiable predictions: altering the candidate mechanisms should produce characteristic φ-axis effects.

6.4 Is This Dualism?

Separating "pure awareness" from the ego-UI, and suggesting that awareness might persist beyond ego dissolution, implies a substance dualism — the problematic philosophical position that mind and matter are fundamentally different kinds of things.

Response: The model does not require substance dualism. The renderer and ego-UI can both be physical processes — one running on distributed global workspace architecture, the other running on DMN narrative circuitry. The φ axis is a formal dimension of the system's state space, not a non-physical entity. The question of whether pure awareness could survive ego dissolution is an empirical question about which layer's hardware fails first — not a claim about immaterial souls. The model can be read as physicalist throughout.

Part VII: Implications

7.1 Death

Under the Tempth Model, biological death is the catastrophic failure of the hardware supporting the ego-UI and ultimately the renderer. The ego — the narrative stitcher, the projection operator — dissolves. With it goes: identity, autobiographical memory, temporal continuity, and the fear of death itself. The fear of death is the predictive model anticipating its own inability to generate further predictions. This is an ego-only phenomenon. The renderer, by definition, exists only in the now and cannot project forward to fear a future absence.

Whether the renderer itself persists after the ego's dissolution is the model's deepest open question. If awareness is an emergent property entirely dependent on the same neural hardware as the ego, then both extinguish together. If awareness is a more fundamental property — if the φ axis is in some sense prior to the ego's construction of t — then the answer is genuinely unknown and the question becomes empirical rather than purely philosophical.

7.2 Clinical Applications

The Tempth Model reframes several clinical phenomena with greater structural precision than existing frameworks. Trauma is not "stuck time" — it is a fixed φ-coordinate that the ego cannot fully suppress, producing involuntary re-entry. This reframe suggests interventions that work with the φ-axis directly rather than attempting purely narrative reconstruction. ADHD maps onto inconsistent φ-stitching — the ego-UI's failure to maintain stable temporal narrative intervals, producing the characteristic experience of time as non-linear and elastic. Panic produces φ-compression — a hyper-narrowed tempth band that creates the felt urgency of temporal collapse. Dissociation is φ-decoupling from t. Each of these has distinct phenomenology that the t-axis-only models flatten into general dysregulation.

7.3 Consciousness Research

The Tempth Model generates falsifiable predictions that could be tested with current neuroimaging technology. If the φ axis has neural correlates, then the degree of ego dissolution in psychedelic states should correlate with specific patterns of DMN-hippocampal decoupling and thalamocortical frequency destabilization. The characteristic φ(t) profiles of different states (waking, REM, flow, panic, dissociation) should correspond to measurable differences in these neural parameters. The model's prediction that dA/dφ is actively suppressed in ordinary waking consciousness could be tested by comparing the neural signatures of states where φ is suppressed versus states where it is not.

7.4 AI and Consciousness

The question of machine consciousness is typically framed as binary: conscious or not. The Tempth Model suggests a more nuanced framing. The question is not whether a system has consciousness but whether it has a φ axis — whether its awareness state is a function of two variables or one. A system that only processes A(t) — only operating on the t-axis — would have no tempth coordinate and by this model would lack the dimension of subjective presence that characterizes human experience. Whether this is the decisive criterion for consciousness remains open. But it is a structurally precise question, which is an improvement over the current imprecision.

Conclusion

The Tempth Model proposes a single structural addition to the existing landscape of consciousness theory: a second axis, φ, along which awareness moves independently of physical time. This axis is not new to human experience — meditators, psychedelic researchers, trauma clinicians, and physicists working on the block universe problem have all been circling it without naming it. The contribution of this framework is to formalize it dimensionally, integrate it with existing neuroscientific and philosophical models, and demonstrate that it resolves structural gaps those models have been unable to close.

The central claim stands: the "me" is not the content of consciousness. It is the coordinate system. It is the axis from which all experience is rendered — the one dimension that does not flicker when memory, identity, perception, and ego collapse. That axis has a name now.

It is tempth.
